EXPERIMENT-19

19. Wireshark Sniffer for Network Traffic & Analysis

Aim

To capture, monitor, and analyze network traffic using Wireshark in Kali Linux and identify protocols such as NTP, DNS, ICMP, HTTP, TLS, and TCP.

Procedure

1. Start Kali Linux in Oracle VirtualBox and open the terminal.
2. Install Wireshark using the installation command.
3. Open Wireshark and select the active network interface, such as eth0 or enp0s3.
4. Start capturing packets before generating network traffic.
5. Execute the commands for NTP, DNS, ICMP, HTTP, TLS, and TCP in the Kali terminal.
6. Apply the appropriate display filters in Wireshark to analyze each protocol.
7. Stop the capture and save the captured packets.

Code / Commands

Step 1: Install and open Wireshark

```
sudo apt update
sudo apt install wireshark -y
sudo wireshark
```

Step 2: NTP (Network Time Protocol)

Command:

```
sudo ntpdate -q pool.ntp.org
```

Step 3: DNS (Domain Name System)

Command:

nslookup google.com

Step 4: ICMP (Internet Control Message Protocol)

Command:

ping -c 4 8.8.8.8

Step 5: HTTP (Hypertext Transfer Protocol)

Command:

curl -I http://neverssl.com

Step 6: TLS (Transport Layer Security)

Command:

curl -I https://www.google.com

Step 7: TCP (Transmission Control Protocol)

Command:

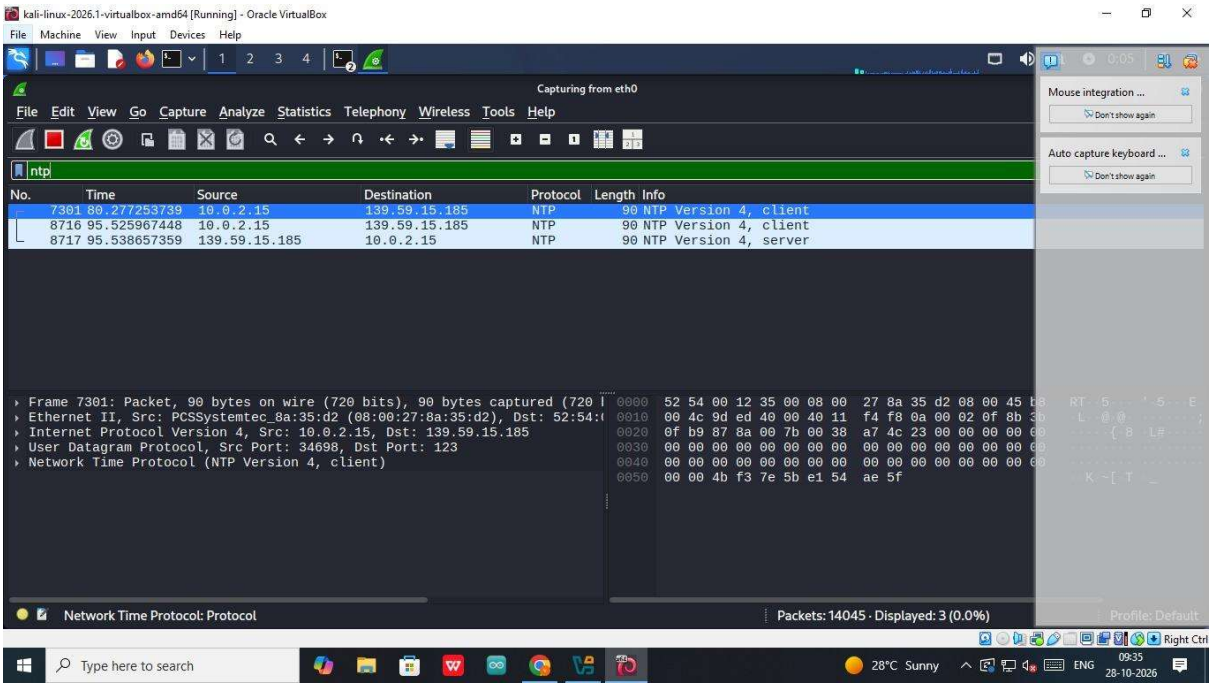
curl -I https://example.com

Sample Input and Sample Output

1.Input:

NTP

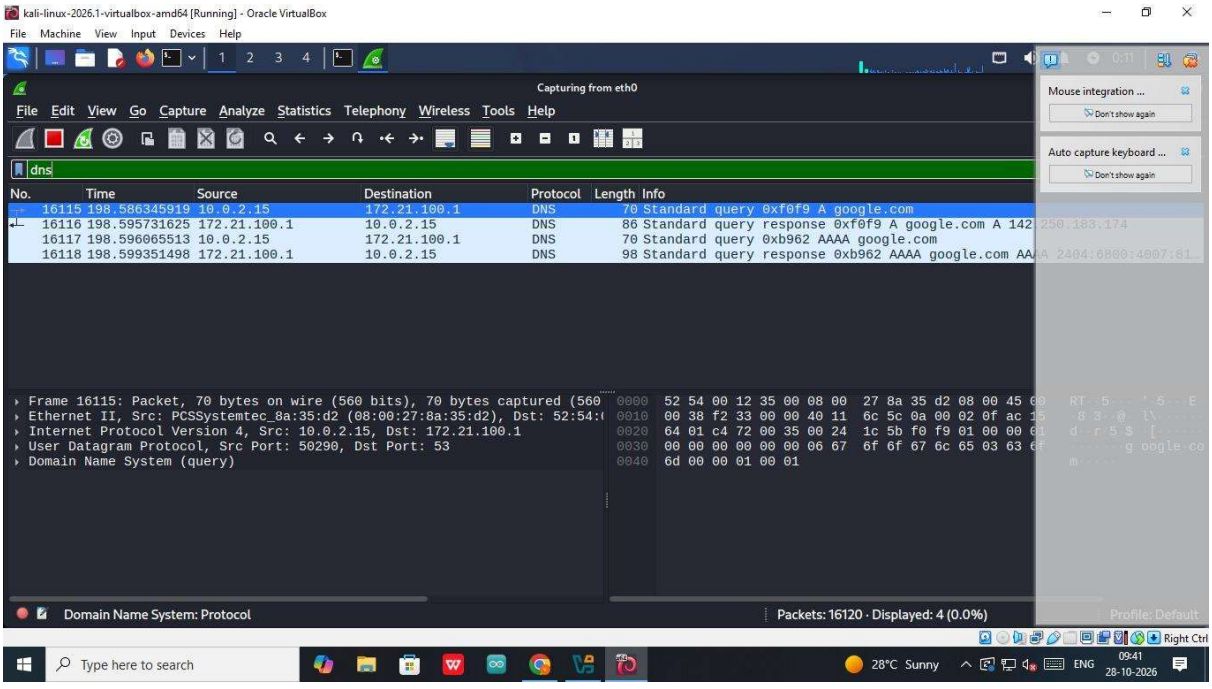
Output:



2.Input:

DNS

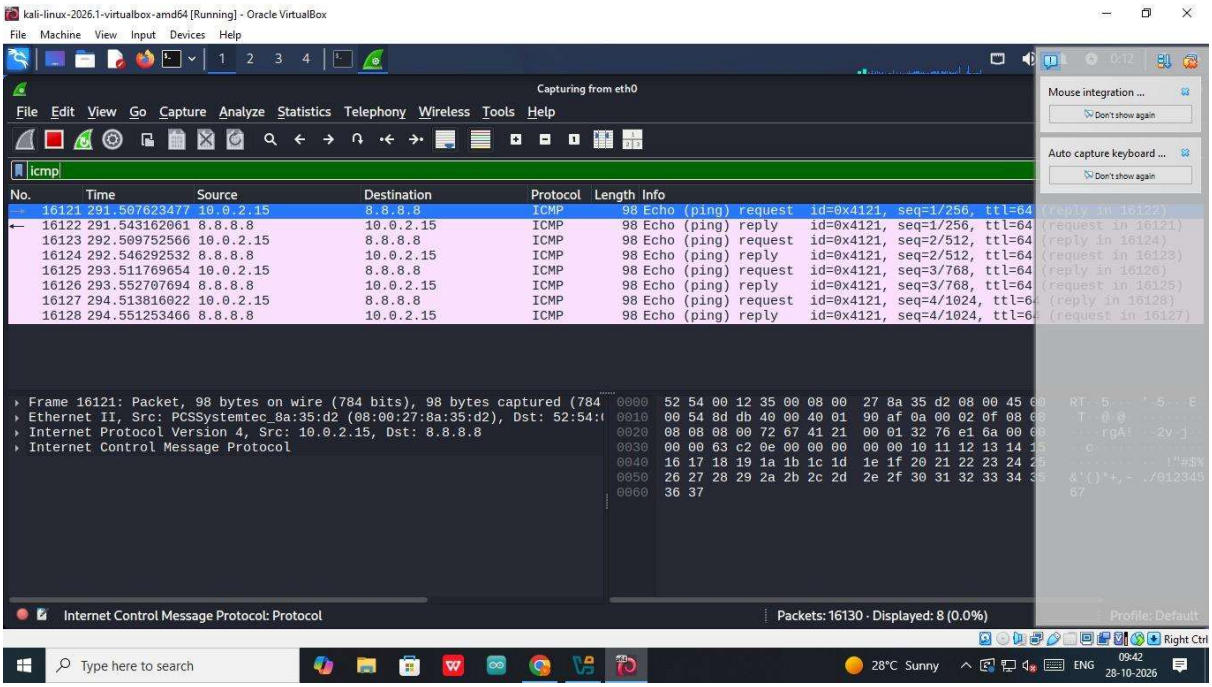
Output:



3.Input:

ICMP

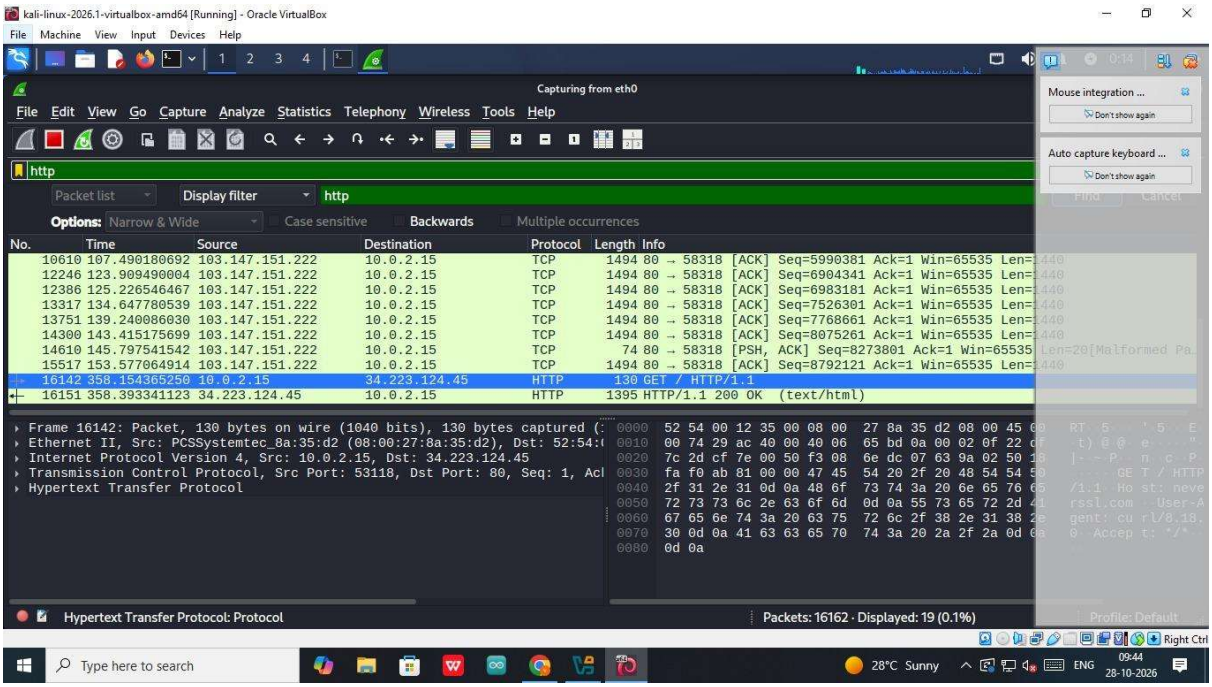
Output:



4.Input:

HTTP

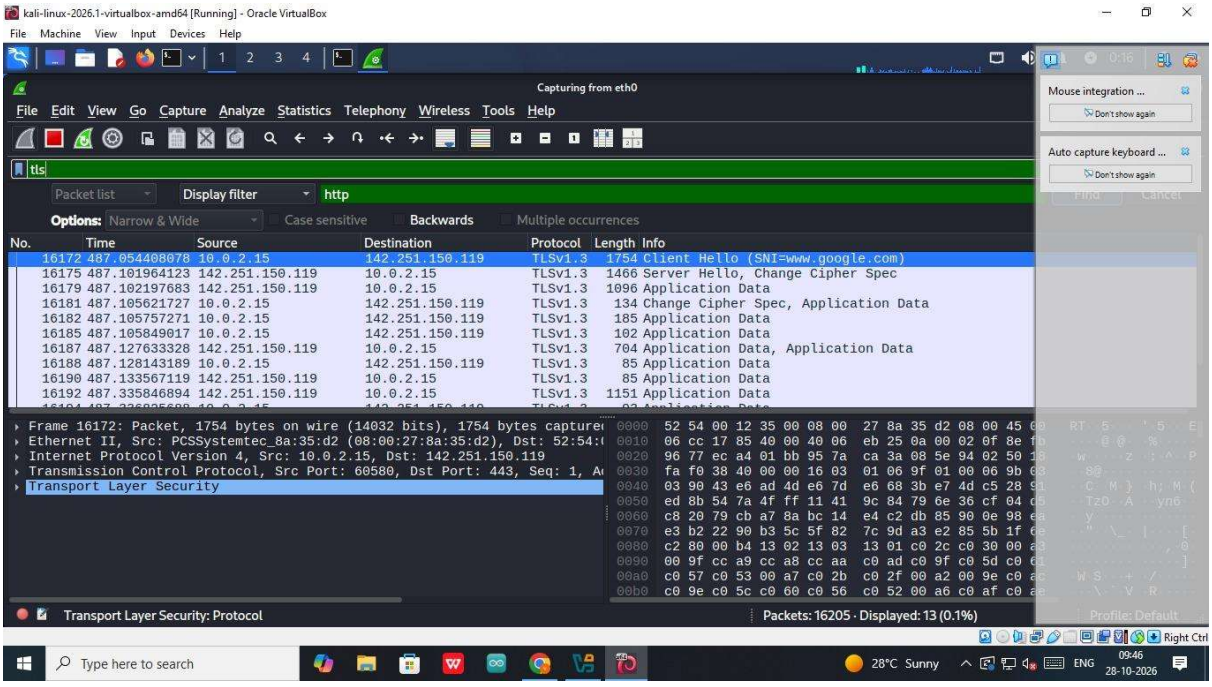
Output:



5.Input:

TLS

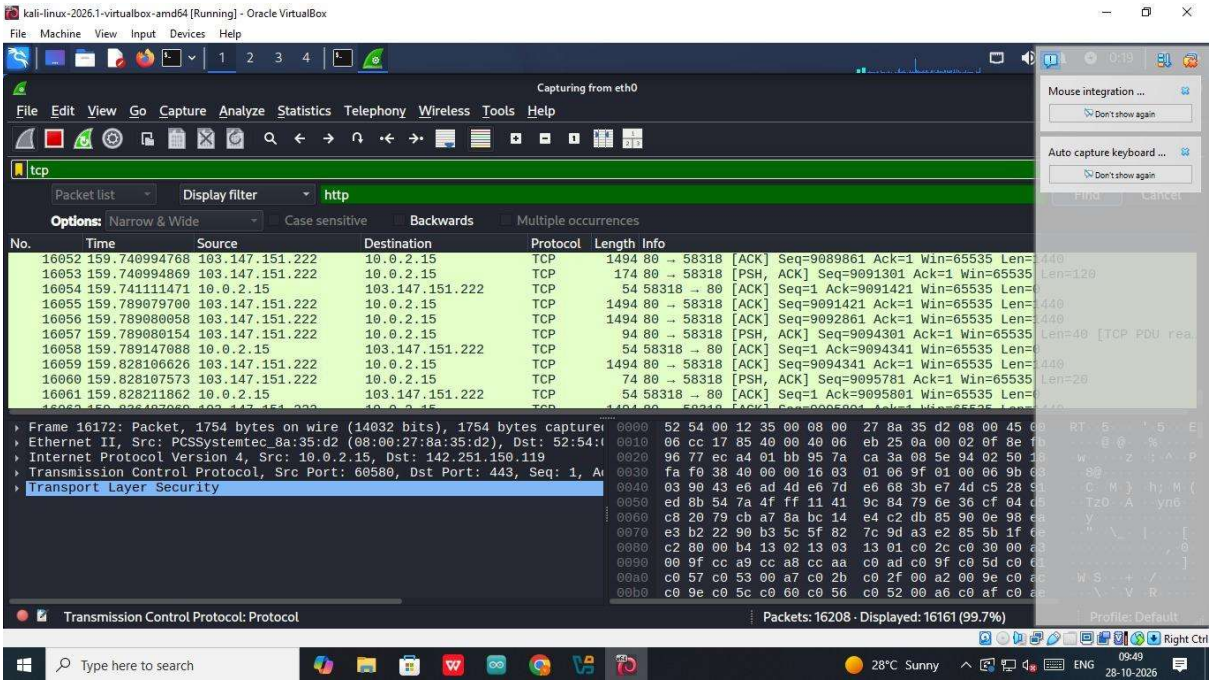
Output:



6.input:

TCP

Output:



Result

Network traffic was generated and analyzed using Wireshark. NTP, DNS, ICMP, HTTP, TLS, and TCP protocols were examined using their respective display filters.